

9.7inch e-Paper (E)

User Manual

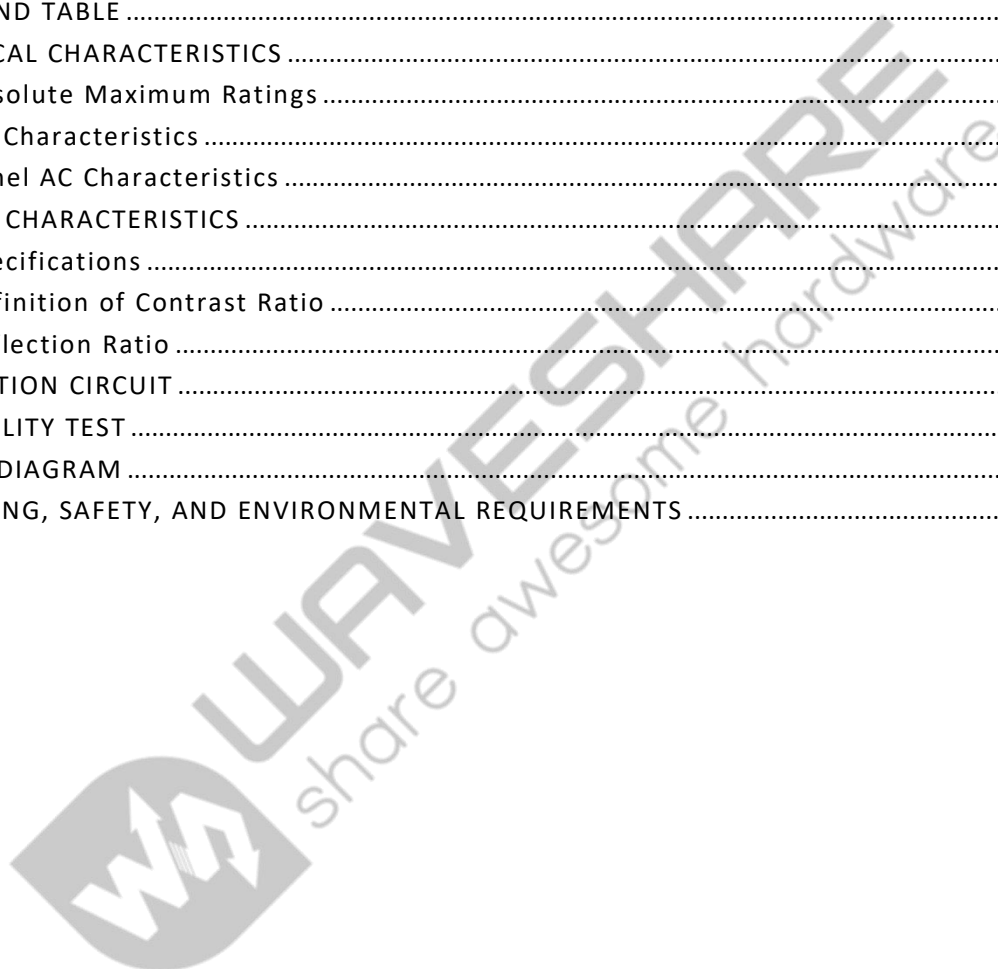


Revision History

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1. OVERVIEW

The display is a reflective electrophoretic E Ink® Spectra™ 6 technology display module built on an active matrix TFT substrate. The panel is capable of displaying black, white, yellow, red, green, and blue images, depending on the associated lookup table used. The circuitry on the panel includes an integrated gate and source drivers, a timing controller, an oscillator, a DC-DC boost circuit, memory for storing the frame buffer and lookup tables, and additional circuitry to control VCOM and BORDER settings.



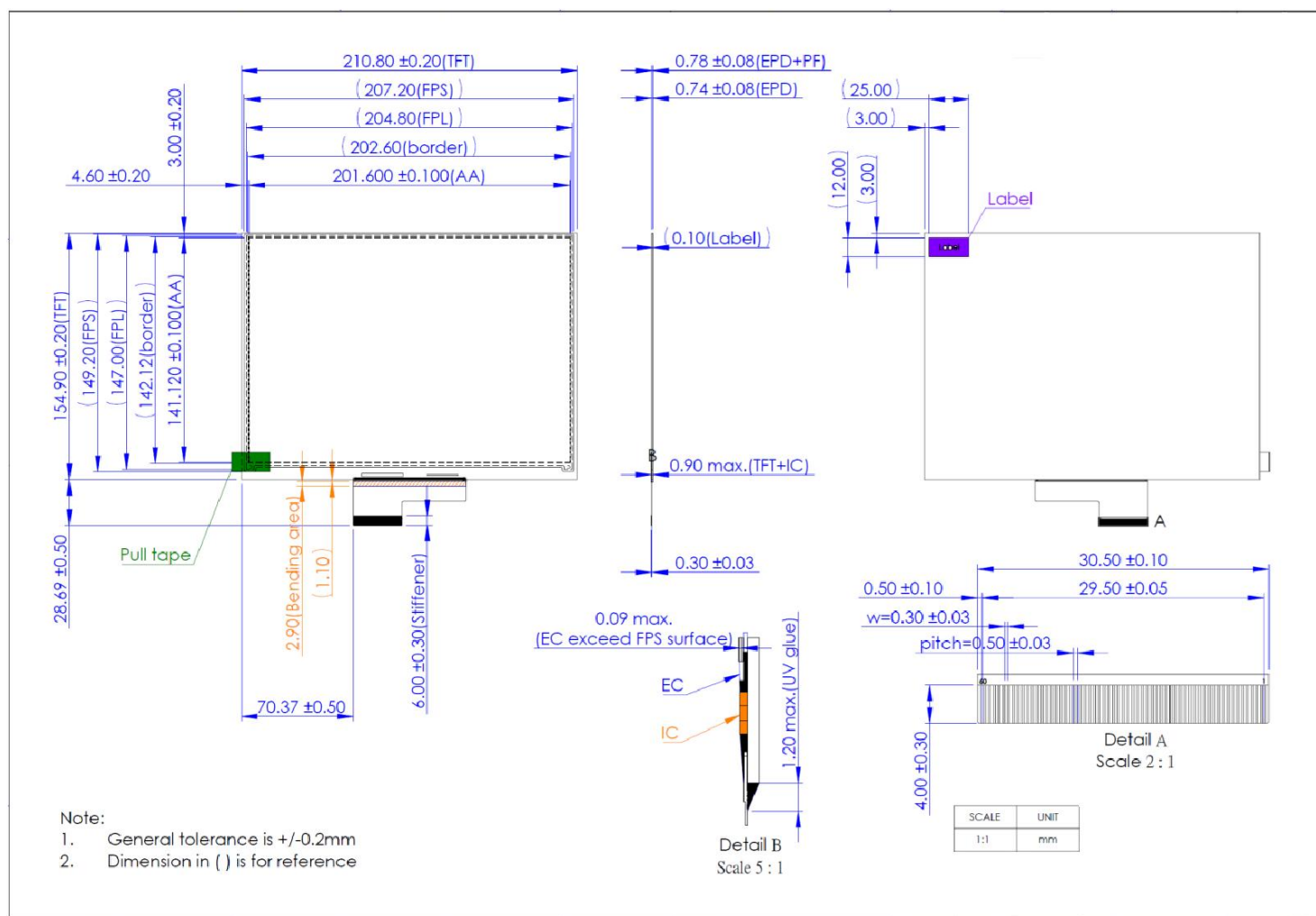
2. FEATURES

- ✧ Highlight Red, Yellow, Green, and Blue colors
- ✧ High contrast
- ✧ High reflectance
- ✧ Ultra-wide viewing angle
- ✧ Ultra-low power consumption
- ✧ Pure reflective mode
- ✧ Bi-stable display
- ✧ Anti-glare hard-coated front surface
- ✧ Low-current deep sleep mode
- ✧ On-chip display RAM
- ✧ Waveform stored in on-chip OTP
- ✧ Serial peripheral interface (SPI) available
- ✧ On-chip oscillator
- ✧ On-chip booster and regulator control for generating VCOM, gate, and source driving voltages
- ✧ I2C master interface for reading external temperature sensor
- ✧ Available in COG package

3. MECHANICAL SPECIFICATIONS

No.	Item	Specifications	Unit	Remark
1	Screen Size	9.7	inch	
2	Display Resolution	1120(H)×784(V)	pixel	141 PPI
3	Active Area	201.6(H)×141.12(V)	mm ²	
4	Pixel Pitch	0.18(H)×0.18(V)	mm ²	
5	Outline Dimension	210.8(W)×154.9(H)×0.74(D)	mm ³	Without MF
6	Module Weight	50±5	g	

4. MECHANICAL DRAWING OF EPD MODULE



5. PIN ASSIGNMENT

5.1 CONNECTOR TYPE AND PIN ASSIGNMENT

Service	Connector	Type Number	Number of Pins	Mating Connector
Interface	P-TWO	196070-60041	60	Copper foil 0.5mm pitch

Pin #	Type	Single	Description	Remark
1	P	VCOMBD_M	VCOMBD driving voltage (Master)	
2	I	RESEC	Current sense input pin for the control loop.(VCOM)	
3	O	GDRC	N-Channel MOSFET gate drive control.(VCOM)	
4	I	RESEN	Current sense input pin for the control loop. (VDDN)	
5	O	GDRN	P-Channel MOSFET gate drive control pin. (VDDN)	
6	I	RESEP	Current sense input pin for the control loop. (VDDP)	
7	O	GDRP	N-Channel MOSFET gate drive control pin. (VDDP)	
8	P	GND	Ground	
9	P	AVDD	Analog voltage supply.(DC/DC)	
10	P	VDD	Analog /digital voltage supply.	
11	--	NC	No connection and do not connect with other NC pins	
12	P	VCC	LDO output pin.	
13	P	VDDP	Positive power supply for analog circuit. (+16V ~ +19V)	
14				
15	--	NC	No connection and do not connect with other NC pins	
16	P	VDDN	Negative power supply for analog circuit. (-19V ~ -16V)	
17				
18	P	VCC1	Positive power supply only for Oscillator.	
19	P	VDDIO	Power input for IO.	
20	I/O	TSCL	I2C Interface to digital temperature sensor Clock pin	
21	I/O	TSDA	I2C Interface to digital temperature sensor Data pin	
22	I	BS0	Input interface setting. Select 3 wire/ 4 wire/ Quad SPI interface.(Default :H)	
23	I	BS1	Input interface setting. Select 3 wire/ 4 wire/ Quad SPI interface.(Default :H)	
24	I	RES#	Reset.	

Pin #	Type	Single	Description	Remark
25	O	BUSY_N	This pin indicates the driver status. Connect with a pull up resistor to VDDIO. BUSY_N = "0" : Driver is busy, data/VCOM is transforming. BUSY_N = "1" : Non-busy. Host side can send command/data to driver.	
26	I	D/C#	Command/Data input. L: command H: data. (4-wire SPI). Connect to GND in 3-wire mode or standard 4-wire mode.	
27	I	CSB_M	Serial communication chip select.(Master)	
28	I	SCL	Serial communication clock input.	
29	I/O	SI0	Serial communication data input/output (3-wire/4-wire SPI). Serial communication data input (Standard 4-wire SPI).	
30	I/O	SI1	Serial communication data input. Serial communication data output. (Standard 4-wire SPI).	
31	I	SI2	Serial communication data input. (Standard 4-wire SPI Quad mode)	
32	I	SI3	Serial communication data input. (Standard 4-wire SPI Quad mode)	
33	O	DRV_P_Gate	Driving external N-MOSFET(VGP)	
34	I	FBP	Positive charge pump(VGP) feedback pin.	
35	P	GND	Ground.	
36	--	NC	No connection and do not connect with other NC pins	
37	O	DRV_N	Driving external BTJ. (VGN)	
38	I	FBN	Negative charge pump feedback pin. (VGN)	
39	O	REG_VGN	VGN internal reference voltage output.	
40	O	DRV_N2	Driving external BTJ. (VNCP_3P5V)	
41	P	VNCP_3P5V	Negative power supply for analog bias1 circuit. (for IFT_Vcom) -3.5V deviation +/- 4.5%	
42	P	VSPH	Positive source buffer output.	
43	P	VSPL	Positive source buffer output.	
44	P	VSPL2	Positive source buffer output.	
45	P	VSPHI	Positive source voltage.	
46	P	VSPLI	Positive source voltage.	
47	P	VSPL2I	Positive source voltage.	
48	P	VSNH	Negative source buffer output.	

Pin #	Type	Single	Description	Remark
49	P	VSNL	Negative source buffer output.	
50	P	VSNL2	Negative source buffer output.	
51	P	VSNL2I	Negative source voltage.	
52	P	VSNLI	Negative source voltage.	
53	P	VSNHI	Negative source voltage.	
54	O	FPL_VCOM	FPL_VCOM driving voltage.	
55	O	TFT_VCOM	TFT_VCOM driving voltage.	
56	P	VBB_3P5V	Negative power supply for analog bias2 circuit. (for TFT_Vcom) -3.5V deviation +/- 4.5%	
57	--	NC	No connection and do not connect with other NC pins	
58	--	NC	No connection and do not connect with other NC pins	
59	P	VGH	Positive Gate driving voltage	
60	P	VGL	Negative Gate driving voltage	

Note 5-1: This pin (CSB) is the chip select input connecting to the MCU. The chip is enabled for MCU communication only when CS# is pulled Low.

Note 5-2: This pin (D/C#) is Data/Command control pin connecting to the MCU. When the pin is pulled HIGH, the data will be interpreted as data. When the pin is pulled Low, the data will be interpreted as command.

Note 5-3: This pin (RES#) is reset signal input. The Reset is active Low.

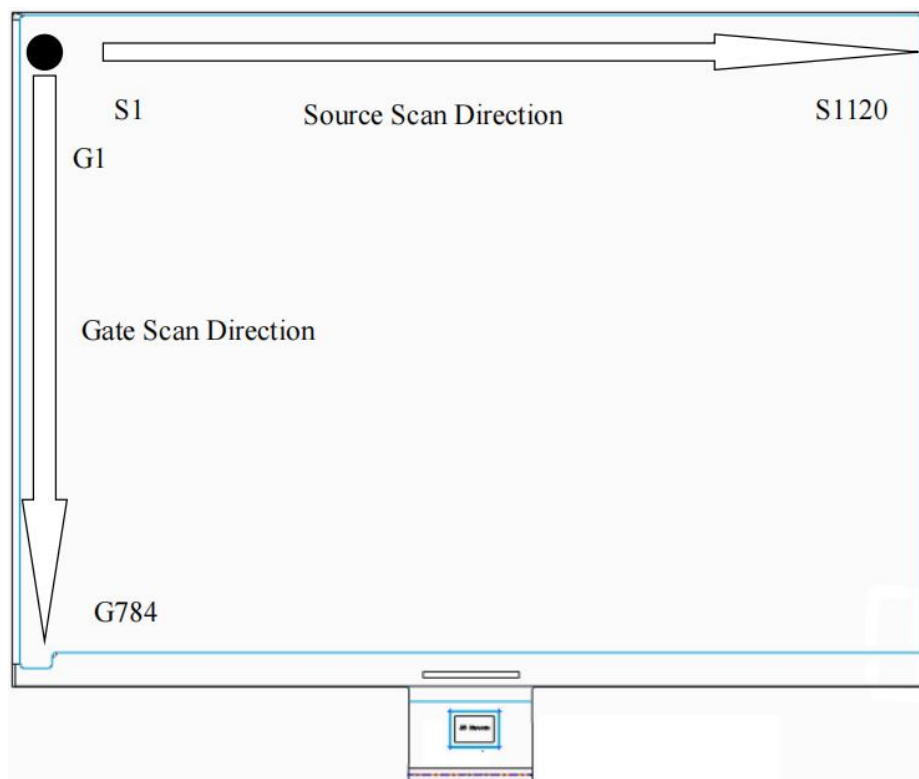
Note 5-4: This pin (BUSY_N) is Busy state output pin. When Busy is low, the operation of chip should not be interrupted and any commands should not be issued to the module. The driver IC will put Busy pin low when the driver IC is working such as:

- Outputting display waveform; or
- Programming with OTP
- Communicating with digital temperature sensor

Note 5-5: This pin (BS0/BS1) is for 3-line SPI /4-line SPI/QSPI selection. Please refer to below Table.

BS1	BS0	MPU Interface
L	L	3-lines serial peripheral interface (SPI) – 9 bits SPI
L	H	4-lines serial peripheral interface (SPI)
H	L	Standard 4-wire SPI
H	H	Standard 4-wire SPI (Default)

5.2 PANEL SCAN DIRECTION



6. COMMAND TABLE

W/R: 0: Write cycle 1: Read cycle C/D: 0: Command 1: Data D7~D0: -: Don't care

	Command	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0	Registers	Setting
1	Power OFF	0	0	0	0	0	0	0	0	1	0		02h
		0	1	0	0	0	0	0	0	0	0		00h
2	Power ON	0	0	0	0	0	0	0	1	0	0		04h
3	Deep Sleep	0	0	0	0	0	0	0	1	1	1		07h
		0	1	1	0	1	0	0	1	0	1		A5h
4	Data Start transmission	0	0	0	0	0	1	0	0	0	0		10h
		0	1	#	#	#	#	#	#	#	#		--
		0	1	..	..	..	..	..	..	..	..		--
		0	1	#	#	#	#	#	#	#	#		--
5	Data Refresh	0	0	0	0	0	1	0	0	1	0		12h
		0	1	0	0	0	0	0	0	0	0		00h

(1) Power OFF (R02H)

Action	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Power OFF	0	0	0	0	0	0	0	0	1	0
	0	1	0	0	0	0	0	0	0	0

(2) Power ON (R04H)

Action	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Power ON	0	0	0	0	0	0	0	1	0	0

(3) Deep Sleep (R07H)

Action	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Deep Sleep	0	0	0	0	0	0	0	1	1	1
	0	1	1	0	1	0	0	1	0	1

Remark: To Exit Deep Sleep mode, User required to send HWRESET to the driver.

(4) Data Start transmission (R10H)

Action	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Data Start transmission	0	0	0	0	0	1	0	0	0	0
	0	1	#	#	#	#	#	#	#	#
	0	1	..	..	..	..	..	..	..	..
	0	1	#	#	#	#	#	#	#	#

After this command, data entries will be written into the RAM until another command is written.

(5) Data Refresh (R12H)

Action	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Data Refresh	0	0	0	0	0	1	0	0	1	0
	0	1	0	0	0	0	0	0	0	0

When this command is received, IC will start the refresh process. BUSY_N will become "0". After the refresh process is finished, BUSY_N will become "1".



7. ELECTRICAL CHARACTERISTICS

7.1 ABSOLUTE MAXIMUM RATINGS

Parameter	Symbol	Rating	Unit
Analog power	VDD	-0.5 to +3.6	V
Operating Temp. range	T _{OPR}	0 to +50	°C
Storage Temp. range	T _{STG}	-25 to +60	°C

Note: Maximum ratings are those values beyond which damages to the device may occur.
 Functional operation should be restricted to the limits in the Electrical Characteristics chapter.

7.2 DC CHARACTERISTICS

Display Module DC characteristics: Please follow the table for the normal operation of the panel; otherwise, it may influence the panel's optical performance.

The standby power (P_{STBY}) is the consumed power when the panel controller is in standby mode. This value is only for reference since it depends on the performance of the panel controller (Whether there is wireless transmission function (WiFi & BLE) will seriously affect the power consumption.)

The following specifications apply for: V_{DD}=3.3V, T_{OPR}=25°C.

DIGITAL DC CHARACTERISTICS						
Symbol	Parameter	Conditions	MIN.	TYP.	MAX.	Unit
VDD	Logic supply voltage		2.4	3.3	3.6	V
VDDIO	IO supply voltage		2.4	3.3	3.6	V
AVDD	DCDC power input voltage		2.4	3.3	6.0	V
VCC	LDO output		1.2	1.3	1.4	V
VCC1	Oscillator supply voltage		1.2	1.3	1.4	V
VGH	Positive Gate driving voltage		21.0	20.0	19.0	V
VGL	Negative Gate driving voltage		-21.0	-20.0	-19.0	V
VSPH	Positive source buffer output		--	Adjusted	--	V
VSPL	Positive source buffer output		--	Adjusted	--	V
VSPL2	Positive source buffer output		--	Adjusted	--	V

VSPHI	Positive source voltage.		--	Adjusted	--	V
VSPLI	Positive source voltage.		--	Adjusted	--	V
VSPL2I	Positive source voltage.		--	Adjusted	--	V
VSNH	Negative source buffer output		--	Adjusted	--	V
VSNL	Negative source buffer output		--	Adjusted	--	V
VSNL2	Negative source buffer output		--	Adjusted	--	V
VSNL2I	Negative source voltage		--	Adjusted	--	V

VSNLI	Negative source voltage		--	Adjusted	--	V
VSNHI	Negative source voltage		--	Adjusted	--	V
VCOM_DC	VCOM_DC output voltage		-4.0	Adjusted	-0.3	V
VIL	Low level input voltage	Digital input pins	GND	--	0.2xVDD	V
VIH	High level input voltage	Digital input pins	0.8xVDD	--	VDD	V
VOH	High level output voltage	Digital output pins, $I_{OUT} = 1\text{ mA}, VDD = 2.4\text{V}$	0.8xVDD	--	--	V
VOL	Low level output voltage	Digital output pins, $I_{OUT} = 1\text{ mA}, VDD = 2.4\text{V}$	GND	--	0.2xVDD	V
IMSTB	Module stand-by current	Stand-by mode	--	150.0	--	uA
IMDS	Module deep sleep current	Deep sleep mode	--	0.7	--	uA
INC	Inrush Current	Booster on	--	683.2	--	mA
P (25°C)	Module operating current	TYP Loading Pattern	--	72.7	97.3	mA
		High Loading Pattern	--	111.0	143.8	mA
	Driving Peak Current	TYP Loading Pattern	--	620.8	866.9	mA
		High Loading Pattern	--	644.8	877.9	mA
	Power Consumption	TYP Loading Pattern	--	240.0	321.1	mW
		High Loading Pattern	--	366.3	474.6	mW
	Image Update time		--	30	--	sec
	Energy Consumption	TYP Loading Pattern	--	2181	--	mAs
		High Loading Pattern	--	3330	--	mAs
PSTBY	Standby Power Dissipation	VDD=3.3V	--	21.8	--	uW

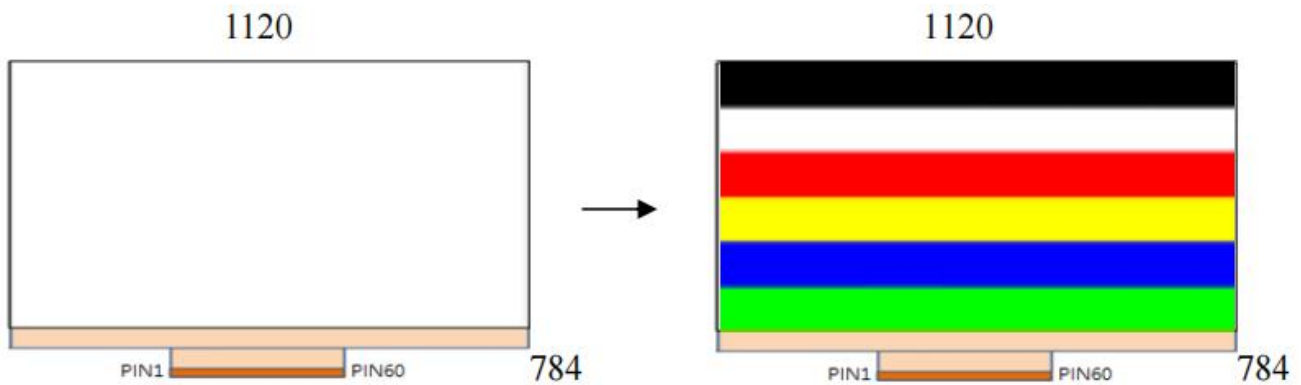
Notes:

1. The Module operating current data is measured by using Oscilloscope, and extract the Mean value.
2. The typical power consumption is measured using associated 25C waveform with following pattern

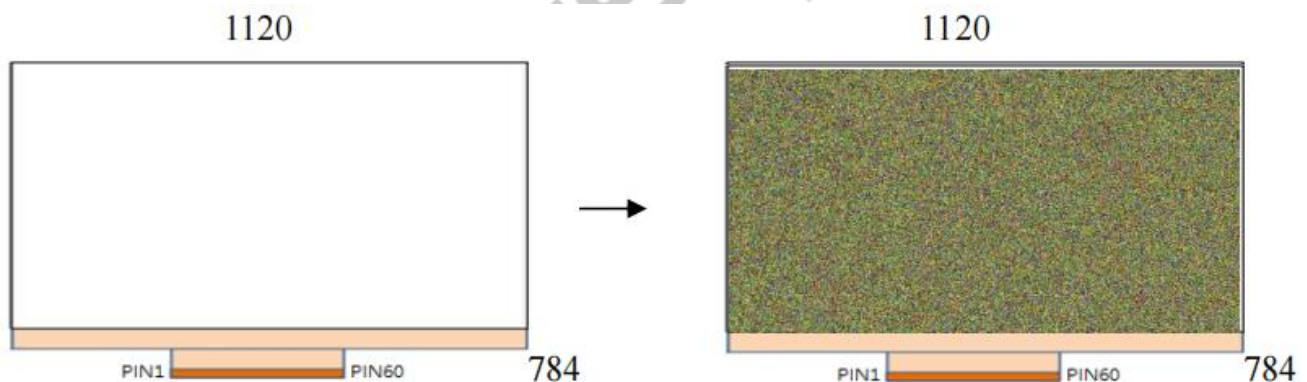
transition: from white pattern to horizontal color stripe pattern. (Note 7-1)

3. The high loading power consumption is measured using associated 25C waveform with following pattern transition: from full white pattern to noise pattern (including random scattering of 6 colors) (Note 7-2)
4. The minimum VDD value by 2.4V is based on typical application pattern with stable and continuing power supply. It does not apply on high loading pattern such as Note 7-2.
5. The listed electrical/optical characteristics are only guaranteed under the controller & waveform provided by E Ink
6. VCOM value has been set in the IC chip on the panel.
7. Due to the presence of the Gate IC, the measured sleep current will increase.
8. Issue the command 0x07 with data 0xA5 to let EPD enter deep sleep mode.
9. Energy Consumption (TYP.) = Module operating current (TYP.) * Image Update time (TYP.)
Energy Consumption (MAX.) = Module operating current (MAX.) * Image Update time (MAX.)

Note 7-1: The typical power consumption



Note 7-2: The high loading power consumption



7.3 PANEL AC CHARACTERISTICS

7.3.1 MCU Interface

7.3.1.1 MCU Interface Selection

In this module, there are 3-wire SPI, 4-wire SPI and standard 4-wire SPI that can communicate with MCU. The MCU interface mode can be set by hardware selection on BS0 and BS1 pins. Please refer to the following table to select the bus interface.

Pin Name	Bus Selection		Data/Command Interface		Control Signal		
	BS0	BS1	SDA	SCL	CS#	D/C#	RES#
3-wire SPI	L	L	SDIN/OUT	SCLK	CS#	N/A	RES#
4-wire SPI	H	L	SDIN/OUT	SCLK	CS#	L : Command Input H : Data Input	RES#
Standard 4-wire SPI	H	H	MOSI : SDOUT MISO : SDIN	SCLK	CS#	N/A	RES#

Table 7-1: MCU interface assignment under different bus interface mode

Note 7-3: L is connected to GND

Note 7-4: H is connected to VDD

7.3.1.2 MCU Serial Interface (4-wire SPI)

The 4-wire SPI consists of serial clock SCLK, serial data SDA, D/C#, CS#.

Function	CS#	D/C#	SCLK
Write Command	L	L	↑
Write data	L	H	↑

Table 7-2: Control pins of 4-wire Serial Peripheral interface

Note 7-5: ↑ stands for rising edge of signal

SDA is shifted into an 8-bit shift register in the order of D7, D6, ... D0. The data byte in the shift register is written to the Graphic Display Data RAM (RAM) or command register in the same clock. Under serial mode, only write operations are allowed.

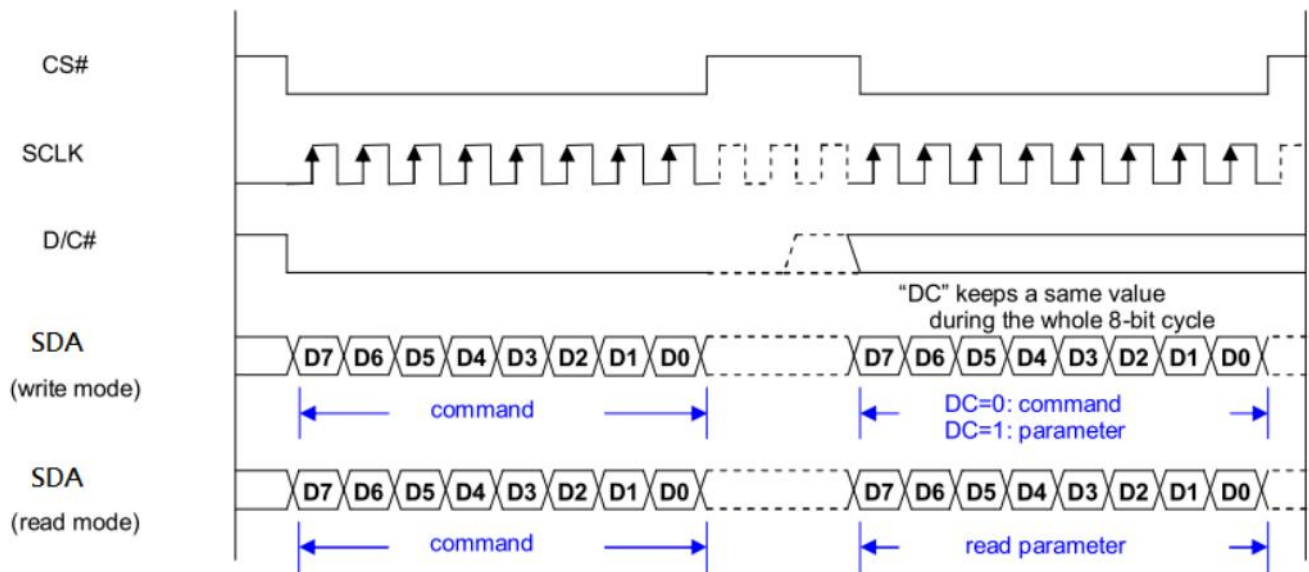


Figure 7-1: Write procedure in 4-wire Serial Peripheral Interface mode

7.3.1.3 MCU Serial Interface (3-wire SPI)

The 3-wire serial interface consists of serial clock SCLK, serial data SDA and CS#. In 3-wire SPI mode, the pin D/C# can be connected to an external ground.

The operation is similar to 4-wire serial interface while D/C# pin is not used. There are altogether 9-bits will be shifted into the shift register on every ninth clock in sequence: D/C# bit, D7 to D0 bit. The D/C# bit (first bit of the sequential data) will determine the following data byte in shift register is written to the Display Data RAM (D/C# bit = 1) or the command register (D/C# bit = 0). Under serial mode, only write operations are allowed.

Function	CS#	D/C#	SCLK
Write Command	L	Tie LOW	↑
Write data	L	Tie LOW	↑

Table 7-3: Control pins of 3-wire Serial Peripheral Interface

Note 7-6: ↑ stands for rising edge of signal

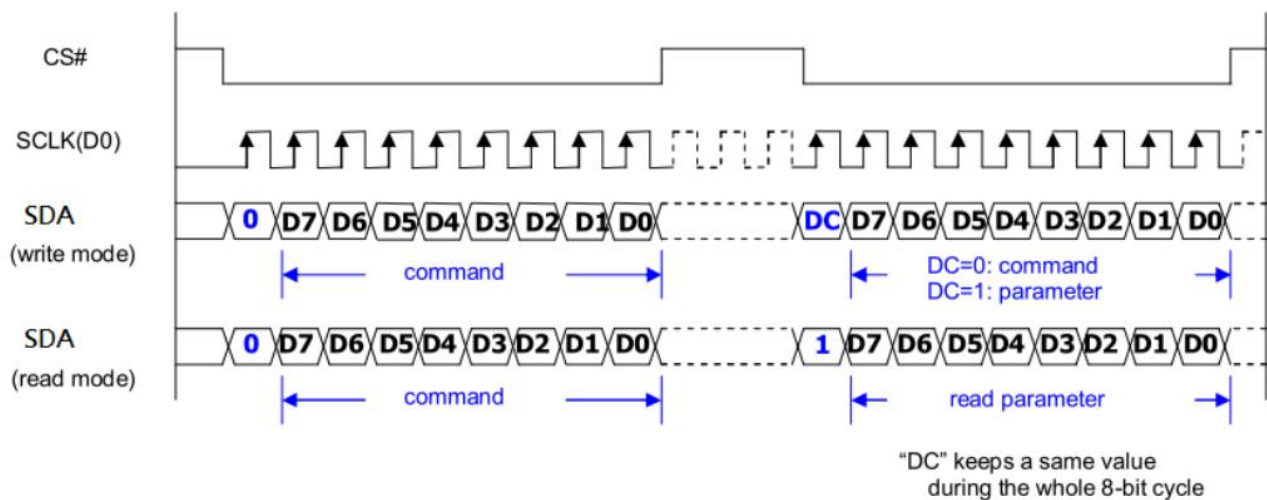
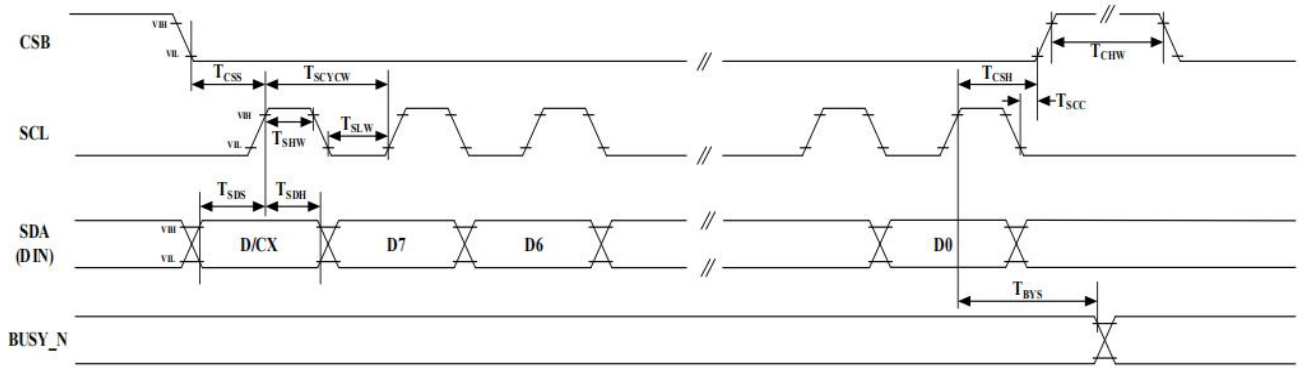


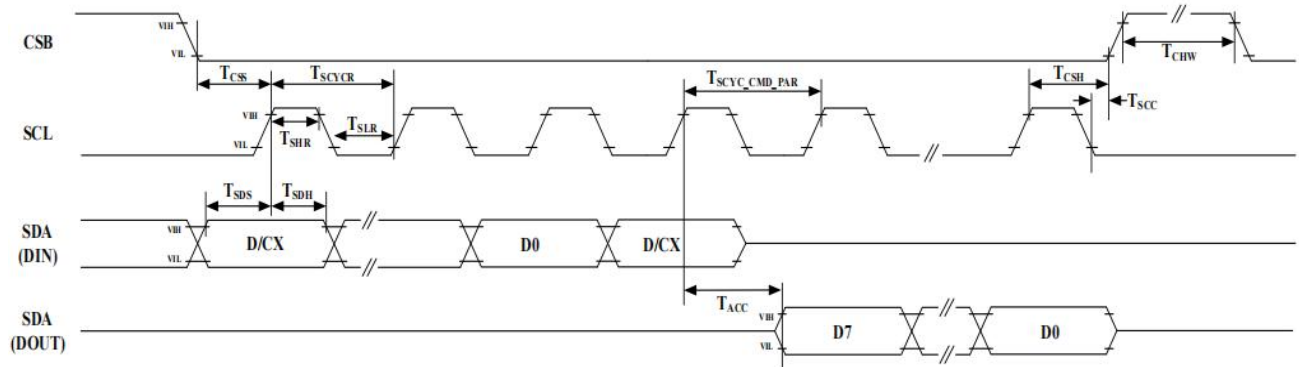
Figure 7-2: Write procedure in 3-wire Serial Peripheral Interface mode

7.3.2 Timing Characteristic of Series Interface

7.3.2.1 3-wire SPI

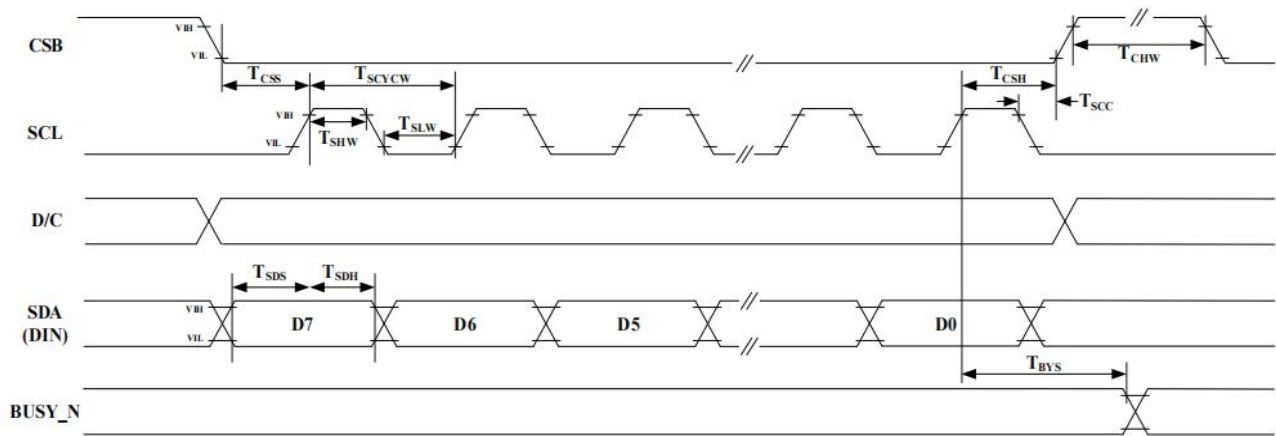


3 pin serial interface characteristics (write mode)

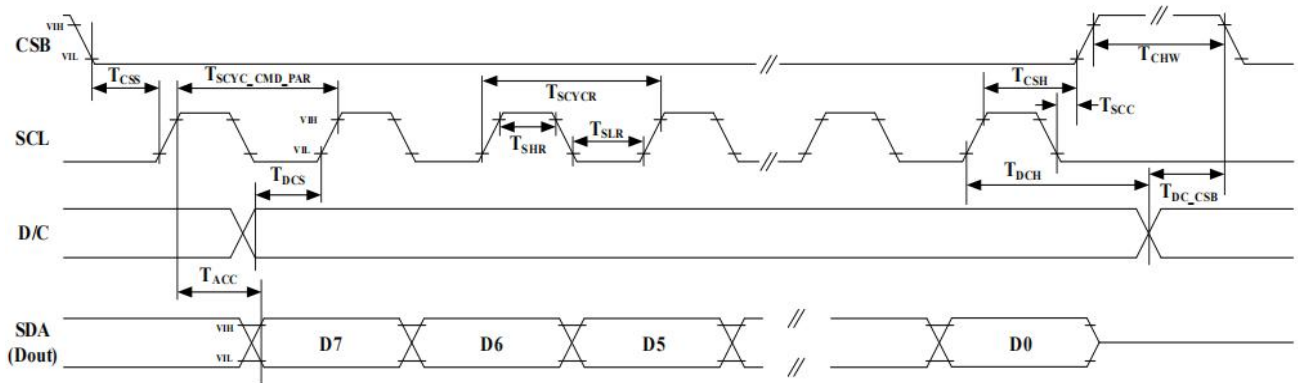


3 pin serial interface characteristics (read mode)

7.3.2.2 4-wire SPI

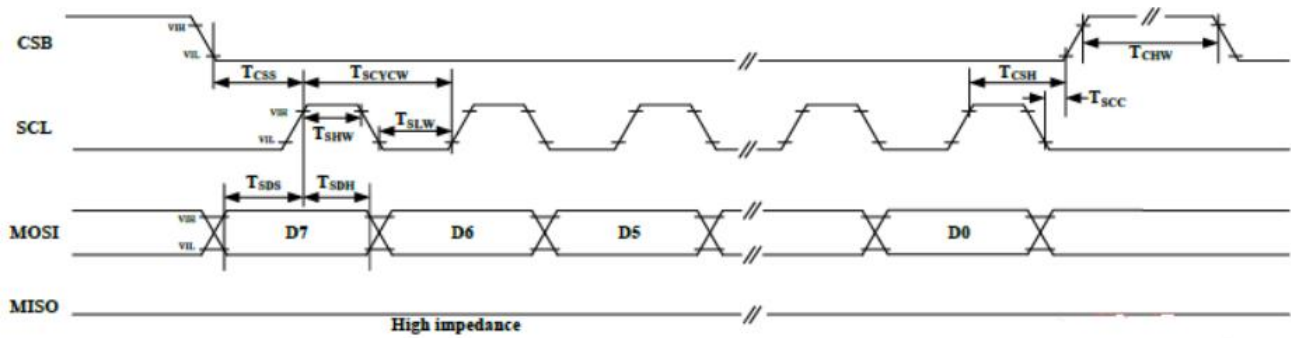


4 pin serial interface characteristics (write mode)



4 pin serial interface characteristics (read mode)

7.3.2.3 Standard 4-wire SPI



Standard 4 pin serial interface characteristics

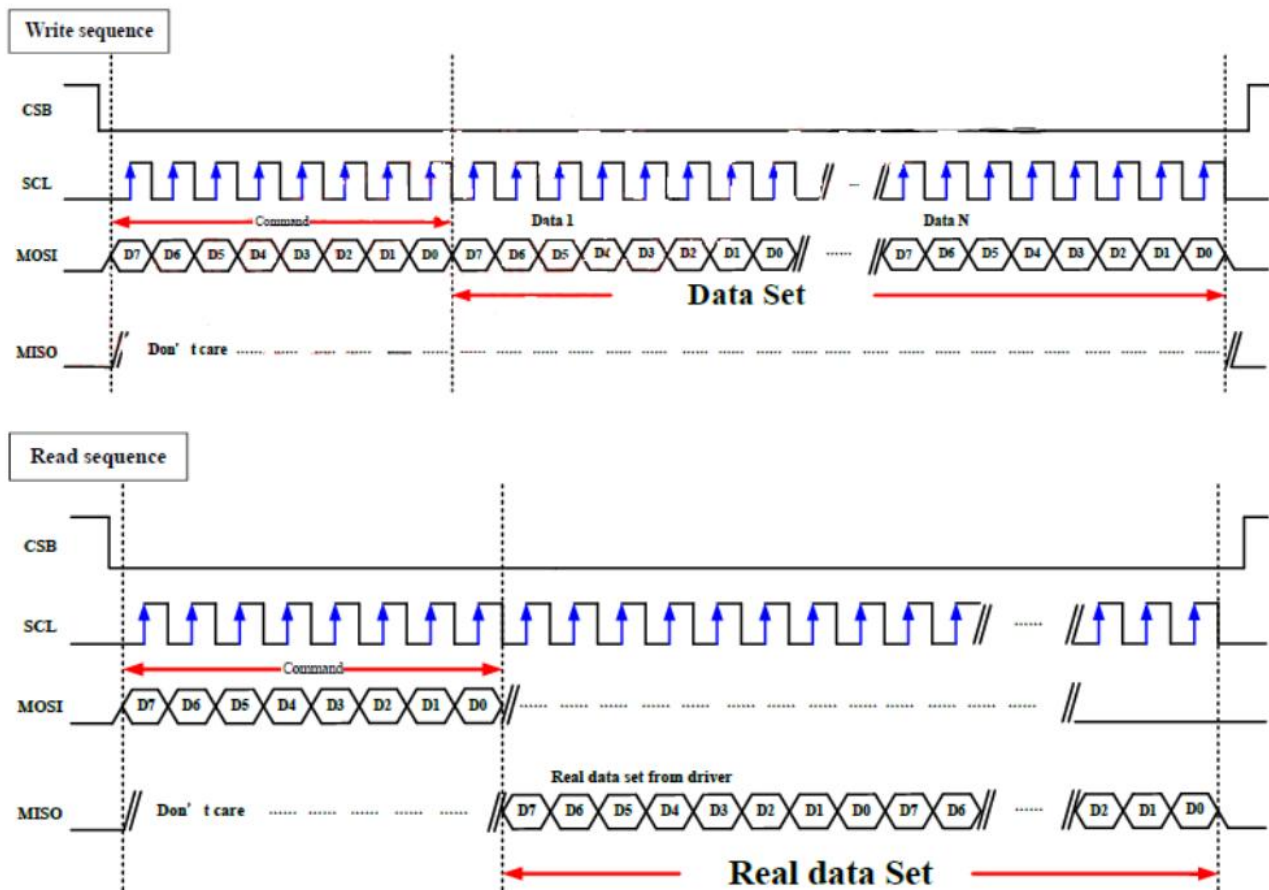


Table 3-wire Timing Table

Parameter	Symbol	Min	Typ	Max	Unit	condition
SERIAL COMMUNICATION						
CSB	Tcss	60			ns	Chip select setup time
	Tcsh	65			ns	Chip select hold time
	Tscc	20			ns	Chip select setup time
	Tchwh	40			ns	Chip select setup time
SCL	Tscycw	50			ns	Serial clock cycle (Write)
	Tshw	25			ns	SCL "H" pulse width (Write)
	Tslw	25			ns	SCL "L" pulse width (Write)
	Tscycr	150			ns	Serial clock cycle (Read)
	Tshr	60			ns	SCL "H" pulse width (Read)
	Tslr	60			ns	SCL "L" pulse width (Read)
	TSCYC_C MD_PAR	150			ns	Serial clock cycle (Between command and 1st parameter)
SDA	Tsds	30			ns	Data setup time
	Tsdh	30			ns	Data hold time
	TACC			140	ns	Access time
BUSY_N	TBYS			150	ns	BUSY_N setup time

Table 4-wire Timing Table

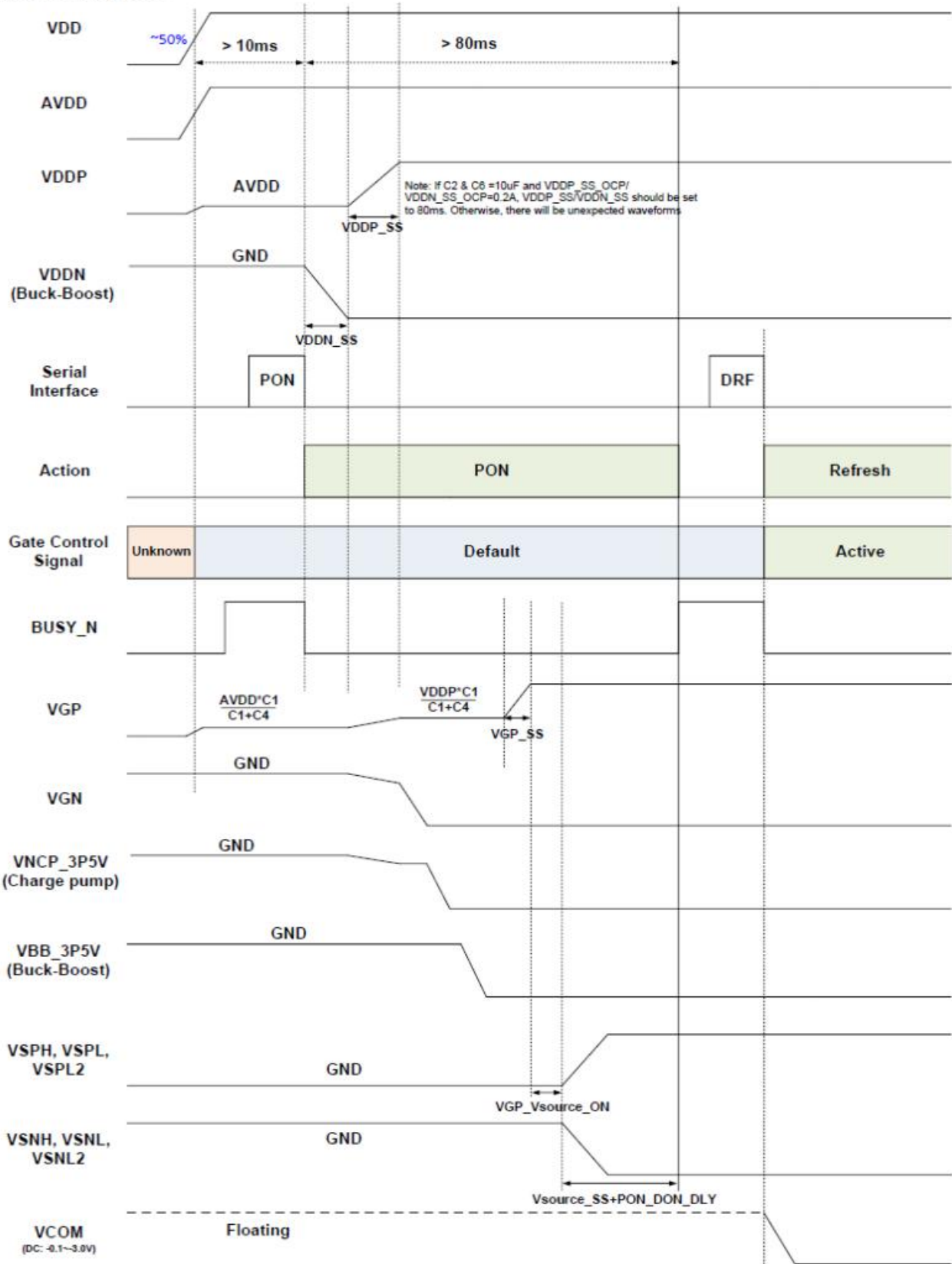
Parameter	Symbol	Min	Typ	Max	Unit	condition
SERIAL COMMUNICATION						
CSB	Tcss	60			ns	Chip select setup time
	Tcsh	65			ns	Chip select hold time
	Tscc	20			ns	Chip select setup time
	Tchwh	40			ns	Chip select setup time
SCL	Tscycw	50			ns	Serial clock cycle (Write)
	Tshw	25			ns	SCL "H" pulse width (Write)
	Tslw	25			ns	SCL "L" pulse width (Write)
	Tscycr	150			ns	Serial clock cycle (Read)
	Tshr	60			ns	SCL "H" pulse width (Read)
	Tslr	60			ns	SCL "L" pulse width (Read)
	TSCYC_C MD_PAR	150			ns	Serial clock cycle (Between command and 1st parameter)
SDA	Tsds	30			ns	Data setup time
	Tsdh	30			ns	Data hold time
	TACC			140	ns	Access time
BUSY_N	TBYS			150	ns	BUSY_N setup time

Table standard 4-wire Timing Table

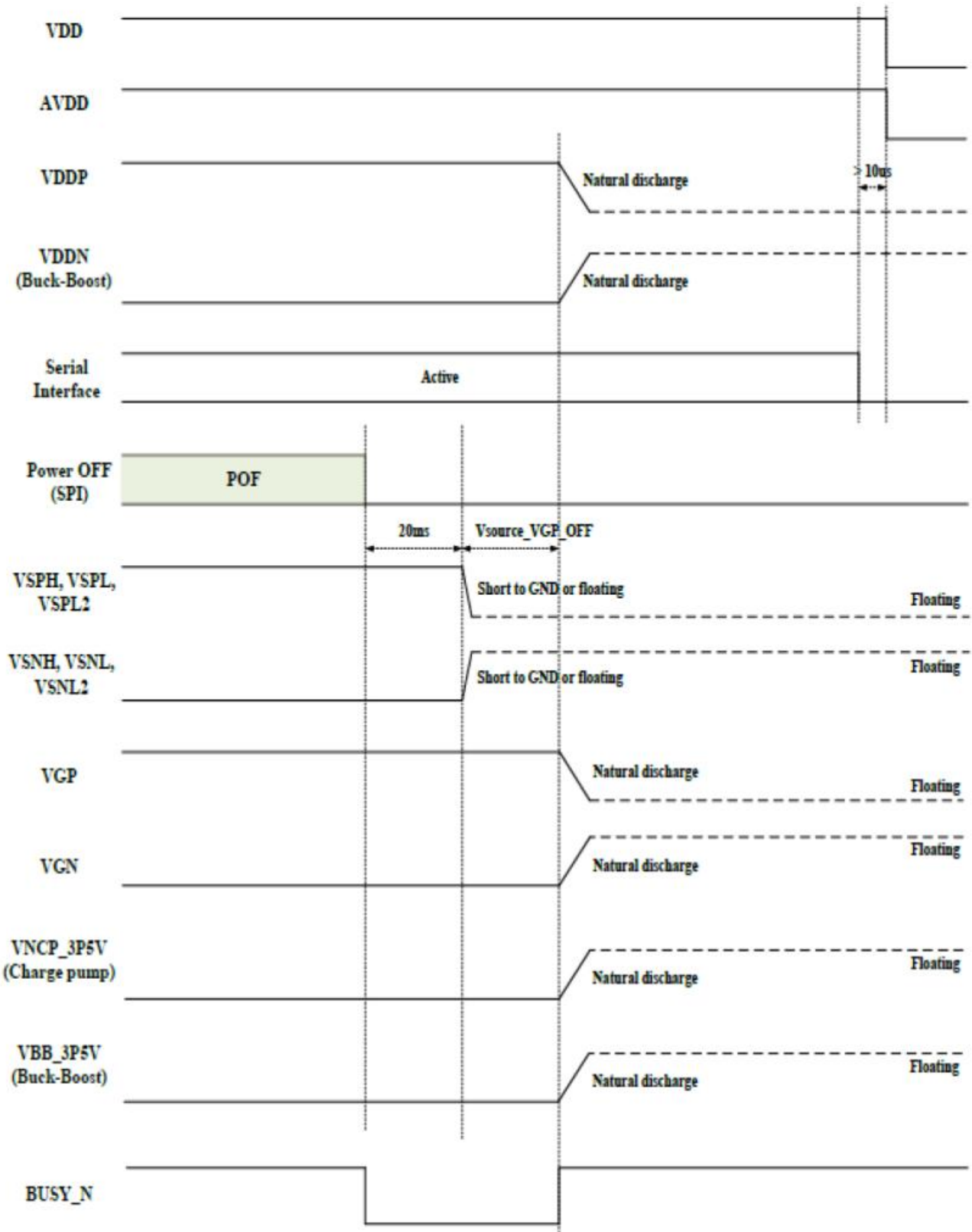
Parameter	Symbol	Min	Typ	Max	Unit	condition
SERIAL COMMUNICATION						
CSB	Tcss	60			ns	Chip select setup time
	Tcsh	65			ns	Chip select hold time
	Tscc	20			ns	Chip select setup time
	Tchwh	40			ns	Chip select setup time
SCL	Tscycw	50			ns	Serial clock cycle (Write)
	Tshw	25			ns	SCL "H" pulse width (Write)
	Tslw	25			ns	SCL "L" pulse width (Write)
	Tscycr	150			ns	Serial clock cycle (Read)
	Tshr	60			ns	SCL "H" pulse width (Read)
	Tslr	60			ns	SCL "L" pulse width (Read)
	TSCYC_C MD_PAR	150			ns	Serial clock cycle (Between command and 1st parameter)
MOSI MISO	Tsds	30			ns	Data setup time
	Tsdh	30			ns	Data hold time
	TACC			140	ns	Access time
BUSY_N	TBYS			150	ns	BUSY_N setup time

7.3.3 Power On/Off Characteristic

Power ON Sequence



Power OFF Sequence



8. OPTICAL CHARACTERISTICS

8.1 SPECIFICATIONS

Measurements are made with that the illumination is under an angle of 45 degrees, the detector is perpendicular unless otherwise specified.

Symbol	Parameter	Conditions	Temperature	Min	Typ.	Max	Unit	Note
T_{update}	Update time	-	0°C	-	45	-	sec	-
Symbol	Parameter	Conditions	Temperature	L* Typ.	a* Typ.	b* Typ.	ΔE_{2000} Max.	Note
WS	White State L*/a*/b* value	White	0°C	65	-4	-2	6	Note 8-1
DS	Dark State L*/a*/b* value	Dark	0°C	13	9	-12	6	Note 8-1
RS	Red State L*/a*/b* value	Red	0°C	26	38	28	6	Note 8-1
YS	Yellow State L*/a*/b* value	Yellow	0°C	65	-13	64	6	Note 8-1
BS	Blue State L*/a*/b* value	Blue	0°C	31	4	-38	6	Note 8-1
GS	Green State L*/a*/b* value	Green	0°C	35	-21	10	8	Note 8-1

Symbol	Parameter	Conditions	Temperature	Min	Typ.	Max	Unit	Note
R	Reflectance	White	25°C	30	34	-	%	Note 8-1
CR	Contrast Ratio	-	25°C	15	22	-	-	-
T_{update}	Update time	-	25°C	-	30	-	sec	-
Symbol	Parameter	Conditions	Temperature	L* Typ.	a* Typ.	b* Typ.	ΔE_{2000} Max.	Note
WS	White State L*/a*/b* value	White	25°C	66	-4	-3	6	Note 8-1
DS	Dark State L*/a*/b* value	Dark	25°C	11	9	-11	6	Note 8-1
RS	Red State L*/a*/b* value	Red	25°C	26	41	30	6	Note 8-1
YS	Yellow State L*/a*/b* value	Yellow	25°C	66	-15	66	6	Note 8-1
BS	Blue State L*/a*/b* value	Blue	25°C	32	4	-39	6	Note 8-1
GS	Green State L*/a*/b* value	Green	25°C	35	-22	10	8	Note 8-1

Symbol	Parameter	Conditions	Temperature	Min	Typ.	Max	Unit	Note
T_{update}	Update time	-	50 °C	-	30	-	sec	-
Symbol	Parameter	Conditions	Temperature	L* Typ.	a* Typ.	b* Typ.	ΔE_{2000} Max.	Note
WS	White State L*/a*/b* value	White	50 °C	66	-3	-2	6	Note 8-1
DS	Dark State L*/a*/b* value	Dark	50 °C	11	10	-12	6	Note 8-1
RS	Red State L*/a*/b* value	Red	50 °C	26	40	30	6	Note 8-1
YS	Yellow State L*/a*/b* value	Yellow	50 °C	66	-15	66	6	Note 8-1
BS	Blue State L*/a*/b* value	Blue	50 °C	32	5	-31	6	Note 8-1
GS	Green State L*/a*/b* value	Green	50 °C	35	-21	5	8	Note 8-1

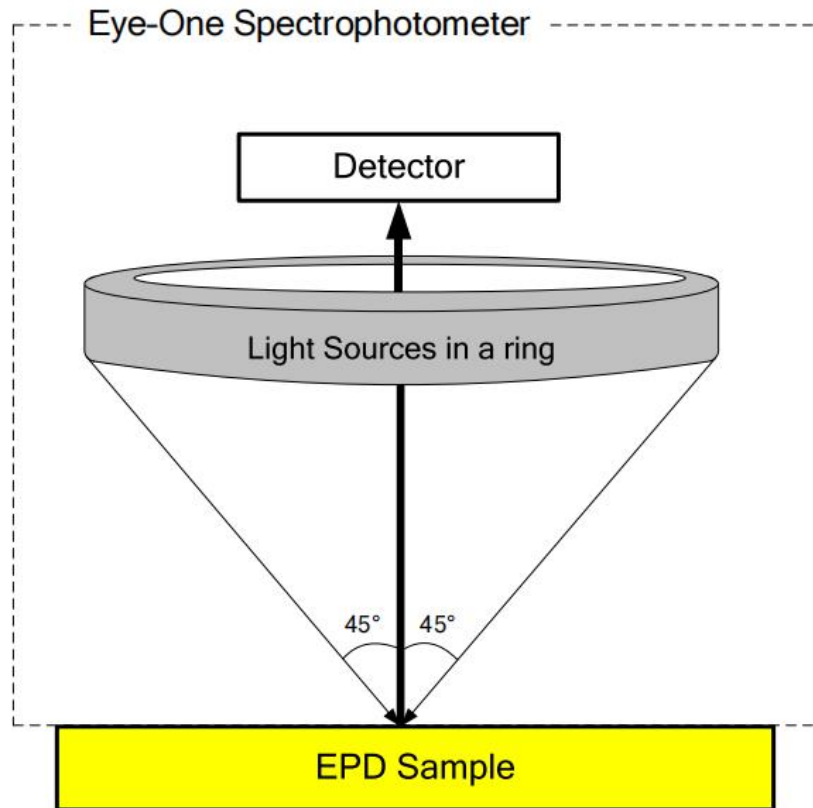
WS: White state, DS: Dark state, RS: Red state, YS: Yellow state, BS: Blue state, GS: Green state.

Note 8-1: Luminance meter: Eye – One Pro3 plus Spectrophotometer

8.2 DEFINITION OF CONTRAST RATIO

The contrast ratio (CR) is the ratio between the reflectance in a full white area (RI) and the reflectance in a dark area (Rd):

$$CR = RI/Rd$$



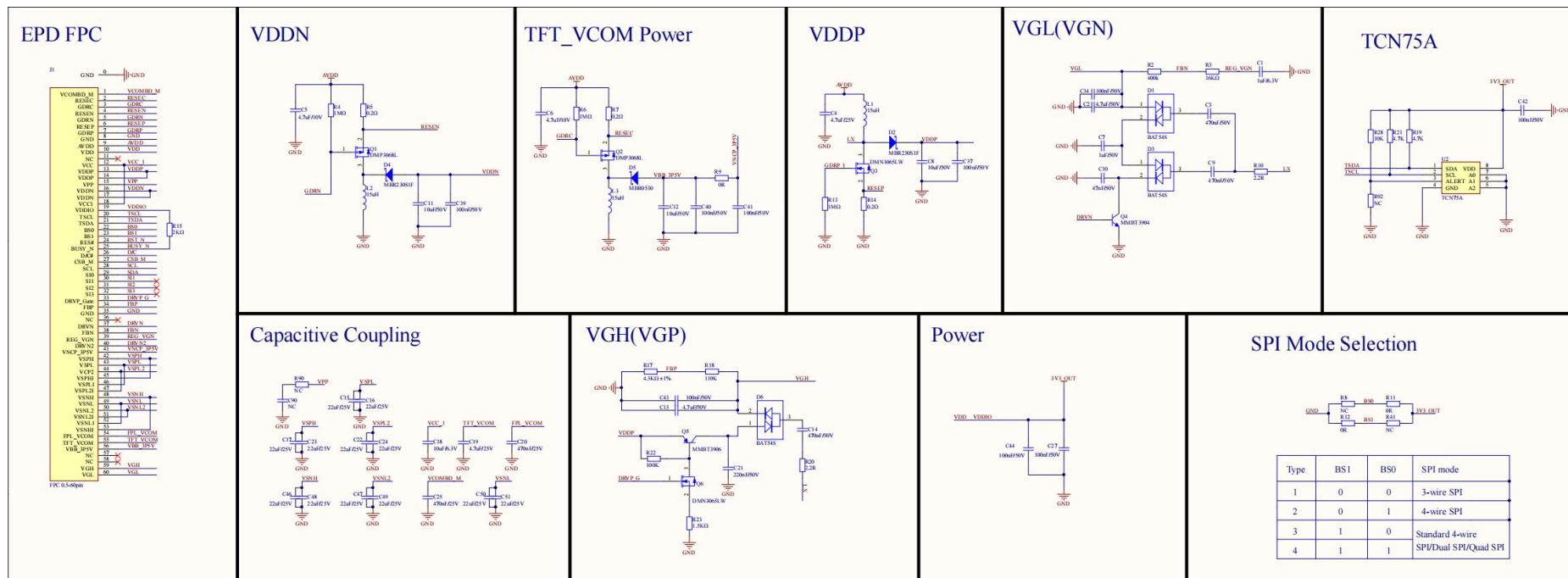
8.3 REFLECTION RATIO

The reflection ratio is expressed as:

$$R = \text{Reflectance Factor}_{\text{white board}} \times (L_{\text{center}} / L_{\text{white board}})$$

L_{center} is the luminance measured at center in a white area ($R=G=B=1$). $L_{\text{white board}}$ is the luminance of a standard white board. Both are measured with equivalent illumination source. The viewing angle shall be no more than 2 degrees.

9. APPLICATION CIRCUIT



10. RELIABILITY TEST

	TEST	CONDITION	METHOD	REMARK
1	High Temperature Storage	Ta= 60°C 40% RH, 240Hrs	IEC 60 068-2-2Bb	(Test in White pattern)
2	Low Temperature Storage	Ta= -25°C, 240Hrs	IEC 60 068-2-1Ab	(Test in White pattern)
3	High Temperature Operation	Ta= 50°C 30% RH, 240Hrs	IEC 60 068-2-2Be	
4	Low Temperature Operation	Ta= 0°C, 240Hrs	IEC 60 068-2-1Ae	
5	High-Temperature, High-Humidity Operation	T = +40°C, RH = 80%, 240Hrs	IEC 60 068-2-78	
6	High Temperature, High-Humidity Storage	Ta= 60°C 80% RH, 240Hrs	IEC 60 068-2-78	(Test in White Pattern)
7	Heat Shock	-25°C(30 min) ~60°C(30 min) 50 cycle, 1Hr/cycle	IEC 068-2-14Nb	(Test in White pattern)
8	Electrostatic Discharge	(Machine model) +/- 200V 0Ω, 200pF	IEC 62180	Non-operation

Actual EMC level to be measured on customer application.

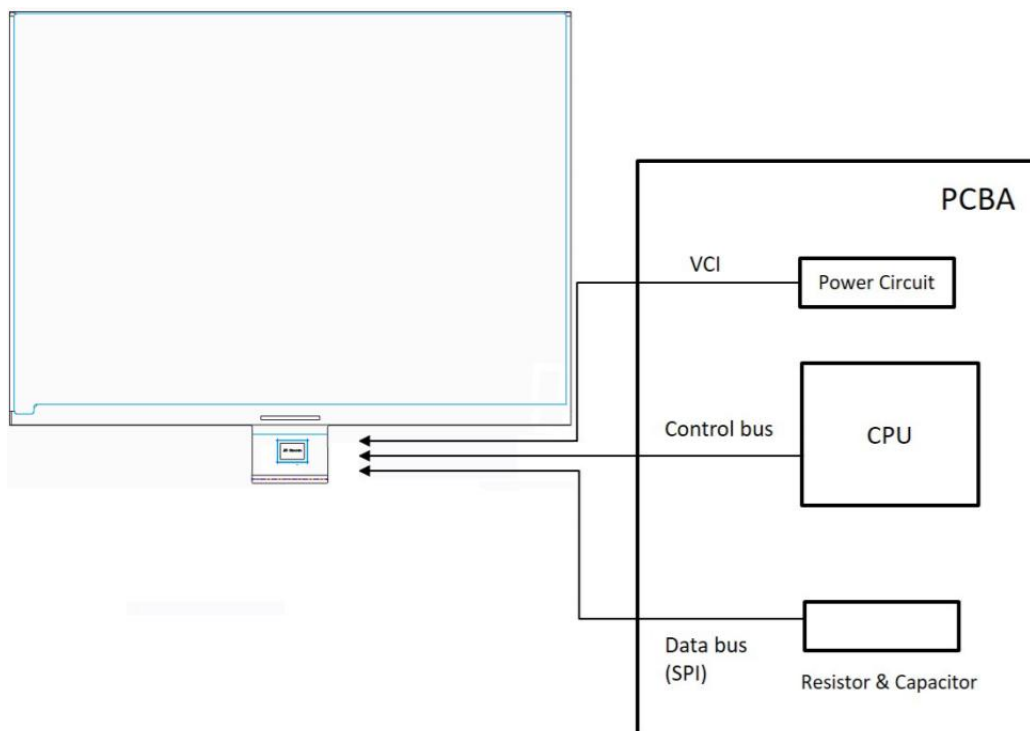
Note: The protective film must be removed before temperature test.

Criteria:

In the standard conditions, there is no display function NG issue occurred.

All the cosmetic specification is judged before the reliability stress.

11. BLOCK DIAGRAM



12. HANDLING, SAFETY, AND ENVIRONMENTAL REQUIREMENTS

WARNING	
The display glass may break when it is dropped or bumped on a hard surface. Handle with care. Should the display break, do not touch the electrophoretic material. In case of contact with electrophoretic material, wash with water and soap.	
CAUTION	
The display module should not be exposed to harmful gases, such as acid and alkali gases, which corrode electronic components.	
Disassembling the display module can cause permanent damage and invalidate the warranty agreements.	
IPA solvent can only be applied on active area and the back of a glass. For the rest part, it is not allowed.	
Mounting Precautions	
(1) It's recommended that you consider the mounting structure so that uneven force (ex. Twisted stress) is not applied to the module.	
(2) It's recommended that you attach a transparent protective plate to the surface in order to protect the EPD. Transparent protective plate should have sufficient strength in order to resist external force.	
(3) You should adopt radiation structure to satisfy the temperature specification.	
(4) Acetic acid type and chlorine type materials for the cover case are not desirable because the former generates corrosive gas of attacking the PS at high temperature and the latter causes circuit break by electro-chemical reaction.	
(5) Do not touch, push or rub the exposed PS with glass, tweezers or anything harder than HB pencil lead. And please do not rub with dust clothes with chemical treatment. Do not touch the surface of PS for bare hand or greasy cloth. (Some cosmetics deteriorate the PS)	
(6) When the surface becomes dusty, please wipe gently with absorbent cotton or other soft materials like chamois soaks with petroleum benzene. Normal-hexane is recommended for cleaning the adhesives used to attach the PS. Do not use acetone, toluene and alcohol because they cause chemical damage to the PS.	
(7) Wipe off saliva or water drops as soon as possible. Their long time contact with PS causes deformations and color fading.	
Data sheet status	
Product specification	This data sheet contains tentative product specifications subjected to changes without notice.
Limiting values	
Limiting values given are in accordance with the Absolute Maximum Rating System (IEC 134). Stress above one or more of the limiting values may cause permanent damage to the device. These are stress ratings only and operation of the device at these or any other conditions above those given in the Characteristics sections of the specification is not implied. Exposure to limiting values for extended periods may affect device reliability.	
Application information	
Where application information is given, it is advisory and does not form part of the specification.	